

Fuad Ahmed

fahmed22@mun.ca | St. John's, NL | LinkedIn | <https://fuad-ahmed.dev/>

Education

Memorial University of Newfoundland
Bachelor of Mechatronics Engineering
Director of Co-op – Engineering Society 'A'

St. John's, NL, Canada
Class of 2028

Experience

Paradigm Engineering

St. John's, NL, Canada

Team Lead & Electrical Designer

September 2023 - Present

- Leading a team of undergraduate students to develop a full-sized electric autonomous go-kart to compete in the Autonomous Karting Series, placing 2nd overall in the 2025 International Grand Prix
- Designed custom power distribution PCBs using buck converters and necessary protections to distribute power from batteries to all onboard motors and sensors.
- Designed a custom STM32-based control PCB to allow the onboard computer to interface with all onboard motors, actuators and remote e-stop, using USB, UART, and CAN peripherals.
- Designed a 10kW drivetrain for an electric vehicle, including batteries, relays, and e-stops.
- Raised over \$45k in project funding through grants and sponsorships.

Formula MUN

St. John's, NL, Canada

High Voltage Systems Lead & Co-Founder

June 2026- Present

- Co-founded the team and led the tractive system design for an FSAE electric race car.
- Designing and assembling a custom 250V battery pack with an 11-point shutoff circuit and BMS.
- Architecting and integrating the inverter and custom drivetrain for a 100 km/h top speed.

Solace Power

St. John's, NL, Canada

Advanced Research Electrical Engineering Intern

May 2026 – August 2026

- Designed, validated, and delivered a customer-facing 400W wireless charging system for autonomous drones operating in harsh environments.
- Modified and tuned MHz frequency resonant power converter and RF power PCBs using VNAs and impedance analyzers for increased efficiency, thermal stability, and signal integrity.
- Researched and modified high voltage wireless power transfer systems using LTSpice and Altium Designer to optimize system efficiency and stability under varying loads
- Board bring up and SMT soldering down to 0402 components for high-density PCBs.

Michelin

Pictou, NS, Canada

Electrical/Automation Engineering Intern

September 2025 – December 2025

- Programmed industrial machinery & HMIs using Allen-Bradley PLC to modernize production lines.
- Produced drawings using SEE Electrical and led installation for a new 2nd-stage tire builder.
- Designed and built a CSA-compliant custom panel to power and control 120VAC and 480VAC motors.

Memorial University Student Design Hub

St. John's, NL, Canada

Innovation & Prototyping Co-op

January 2025 – April 2025

- Completed advanced training on CNC Milling, DFM and CAM profiling.
- Developed manuals and provided instruction for first-year students for FDM 3D Printers.

Skills

Software: KiCad, Altium Designer, Schematic Capture & Layout, LTSpice, Matlab, C++, Python

Electronics: Analog & Digital Circuit Design, PCB Design, Power Electronics, Board Bring-Up, Soldering

Lab Testing: Oscilloscopes, Function Generators, Impedance Analyzers, Vector Network Analyzers